

Investigation of clad ballooning during NSRR RIA tests using ALCYONE fuel performance code

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ARTICLE INFO

Article history:

Received 10 December 2021

Revised 4 February 2022

Accepted 5 February 2022

Available online 6 February 2022

Keywords:

PWR

RIA

NSRR

ALCYONE code

Film boiling duration

Clad ballooning

FGR

Gas axial flow

ABSTRACT

ALCYONE simulations of integral RIA transients performed on UO₂ fuel rods prior to or within the ALPS program in the Japanese NSRR facility with stagnant liquid water coolant conditions are presented in this paper. The 11 selected tests which cover a wide range of fuel enthalpy increases (200 J/g – 800 J/g) aimed at assessing/challenging ALCYONE capabilities on four quantities of interest: clad outer surface peak temperature (293 K – 1200 K), film boiling duration (0 s – 2 s), clad residual hoop strain (0% – 25%) and transient fission gas release (2% – 25%). Despite a global consistency between measurements and ALCYONE predictions, paths are investigated to explain and reduce discrepancies. In particular, the reevaluation of fuel enthalpies by JAEA has led to revisit ALCYONE clad-to-water coolant heat exchange models which were derived from previous values of fuel enthalpies and suggested a recalibration of some of their parameters. Then, modeling the delayed gas axial flow in the free volumes of the rod is shown essential to achieve better residual hoop strain predictions in case of clad ballooning if proper timing of fission gas release, rod internal pressure increase and clad temperature elevation can be simulated. Key experimental and/or modeling research areas are shown to be at stake for future work.

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1. Introduction

Co-developed within the PLEIADES software environment by CEA, EDF and Framatome, ALCYONE is a multidimensional finite element-based nuclear fuel performance code dedicated to the analysis of the behavior of fuel rods during irradiation in commercial Pressurized Water Reactors (PWRs), power ramps in experimental reactors or accidental conditions, including Reactivity-Initiated Accidents (RIAs). Capable of steady state and transient thermal modeling, ALCYONE solves fully-coupled equations of thermo-mechanics together with sophisticated models for fission gas swelling and release [1,2]. With regard to RIA experiments, ALCYONE is capable of modeling the Pellet-Cladding Mechanical Interaction (PCMI) [2,3] and post-Departure from Nucleate Boiling (DNB) phases as well as various fuel/cladding material types. ALCYONE is in particular validated for PWR-UO₂ fuels with advanced claddings under “low pressure-low temperature” or

“high pressure-high temperature” water coolant conditions [4] and for PWR-MOX fuels under sodium coolant conditions [5]. Current development work is focused on tests to be performed on high-burnup fuel rods under prototypical PWR conditions within the CABRI International Program [6] where heat transfer mechanisms related to transient boiling as well as Fission Gas Release (FGR) may induce clad failure by ballooning.

With a view to enlarging ALCYONE RIA validation base on water coolant tests and to test the code capability to reproduce clad ballooning, a selection of integral RIA transients performed on UO₂ fuel rods in the Japanese NSRR facility with stagnant liquid water coolant conditions is considered in this paper. The reevaluation in 2014 of fuel enthalpies performed by the Japan Atomic Energy Agency (JAEA) led to the plot of several measures of interest as a function of the enthalpy increase during the tests [7,8]: the clad outer surface peak temperature, the film boiling duration, the maximum clad residual hoop strain and the transient FGR. Overall, this set of data gives a unique opportunity to challenge ALCYONE models related to clad-to-coolant heat exchanges, FGR and associated clad ballooning. One important aspect of NSRR tests is the often low initial internal pressure of the rodlet which makes clad ballooning unlikely if extensive FGR and local overpressure are not

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Table 1
Main features of the selected integral tests.

Test ID.	Characteristics
GK-1, GK-2	RT capsule Varying initial inner pressure [0.1 MPa – 4.3 MPa]
HBO-2, HBO-3, HBO-4	RT capsule Varying pulse FWHM [4.4 – 6.9 ms] Varying initial inner pressure [0.1 MPa – 5.1 MPa] Very high transient FGR (around 20%)
TK-1, TK-4, TK-5, TK-6, TK-8, TK-9	RT capsule Varying initial inner pressure [0.1 MPa – 3 MPa] Large range of measured clad residual strain [0.3% – 25%]

RT: Room Temperature. FWHM: Full Width at Half Maximum.

considered. The modeling of the high residual clad hoop strains (up to 25% locally) in some of NSRR tests is therefore a challenging task for any fuel performance code. Similar concerns were recently reported in the CSNI RIA synthesis reports [9,10] about the necessary improvement in fuel performance codes of models for clad-to-coolant heat transfer under very fast transients and transient boiling conditions, for fission gas dynamics, e.g. transient release and axial transport, and for cladding high-temperature mechanical behavior under RIA-relevant conditions.

In this paper, the selected NSRR tests to be simulated with ALCYONE are first presented. ALCYONE main features for pulse-irradiation simulation are then recalled with a focus on the modeling of the water boiling curve for RIA transient conditions. The comparison of the results from reference ALCYONE simulations with relevant experimental results (clad outer surface peak temperature and film boiling duration) are then discussed. A recalibration of the boiling curve parameters is proposed to reduce discrepancies between measured and calculated quantities. In the last part, calculated clad residual hoop strain and transient FGR are assessed and their sensitivity to the fuel permeability and to the FGR kinetics discussed.

2. NSRR tests

2.1. Description of the selected NSRR test cases

Among the RIA-simulation tests performed prior to or within the ALPS (Advanced LWR fuel Performance and Safety research) program by JAEA in the NSRR facility, a selection of 11 tests pertaining to the GK, HBO, and TK-series related to PWR fuel rods¹ are considered in this paper. To enlarge ALCYONE validation base, these 11 cases were chosen in order to encompass a wide range of rodlet and test characteristics, as well as measurement and examination results, among all information possible gathered through an up to date literature review [7,8,11–41]. The main features of the selected integral tests are synthesized in Table 1. Compiled details are provided in Appendix.

The fuel segments were extracted from either 17×17 or 14×14 lattice design PWR full-length fuel rods previously irradiated in different Japanese commercial reactors to high burnup levels (38 to 51 GWd/tM is the range of local burnup achieved). They were composed of UO₂ fuel pellets sheathed with standard or low tin Zircaloy-4 claddings. Refabricated with a 110 to 135 mm long pellet stack active length, the test fuel rodlets were pressurized with helium (0.1 to 5.1 MPa). Two types of experimental capsules are used in the NSRR facility, namely the Room Temperature (RT²) test capsule or the High Temperature (HT²) test capsule

¹ The test series are named after the power reactor where the parent fuel rods are irradiated (GK for Genkai, HBO for Ohi till High-Burnup, and TK for Takahama respectively).

² RT and HT stand for « Room-Temperature, Atmospheric-Pressure » and « High-Temperature, High-Pressure » respectively.

both with a stagnant water coolant initially at atmospheric pressure and room temperature or at high pressure (6.4 MPa) and high temperature (280 °C), respectively. The 11 selected tests pertain to RT-type only.

Test pulses had a Full Width at Half Maximum (FWHM) around 4.4 ms for most of them and yielded radial average peak fuel enthalpy increase within a wide range (200 to 800 J/g). Some of the selected tests exhibited departure from nucleate boiling and film boiling during a few seconds leading to a cladding temperature excursion following the power pulse. Tests resulted in a wide range of transient FGR (5% – 26%) determined by rod puncturing and rodlet residual hoop strain (up to 25% with regard to the local maximum). Nevertheless none experienced any clad failure.

2.2. Synthesis of NSRR test measurements

For all the NSRR tests performed before 2014, the calculated peak fuel enthalpy increases have been recently reevaluated by JAEA [7,8]. The reevaluated enthalpies were 10 to 65% higher than the original values. Graphs illustrating the evolution of the following four quantities of interest with the new peak enthalpy increases were provided by JAEA: clad outer surface peak temperature, film boiling duration, clad residual hoop strain and transient fission gas release. The data available for PWR fuel rods only are reproduced in Fig. 1.

The graphs include a large number of tests that exceeds the 11 selected NSRR transients. Considering that the tests conditions are very similar (FWHM in particular), they give interesting trends that may be summarized as follows: peak clad temperature starts to rise after 400 J/g – 500 J/g of enthalpy increase and reaches ~1200 K at 600 J/g – 750 J/g, film boiling time seems to rise at a higher enthalpy level (600 J/g) and reaches a maximum of ~2 s at 750 J/g, the residual clad hoop strain presents two slopes with a slow increase up to 600 J/g, followed by a fast rise afterwards, characteristic of clad ballooning. No clear trend is however seen for FGR when plotted as a function of enthalpy increase, as pointed out in the RIA synthesis report [9]. The very important dispersion is for some part related to the large range of fuel burnups tested in the NSRR facilities.

In the next sections, ALCYONE main models will be described and simulation results will be compared to the measurements gathered in Fig 1.

3. Outline of the fuel performance code ALCYONE

3.1. Main features

ALCYONE pulse-irradiation simulation capability includes (see details in [1,2,4,5]):

- the solving of the transient thermal heat balance equation and of the mechanical equilibrium for the pellet-gap-cladding system, based on the finite element method [42], considering

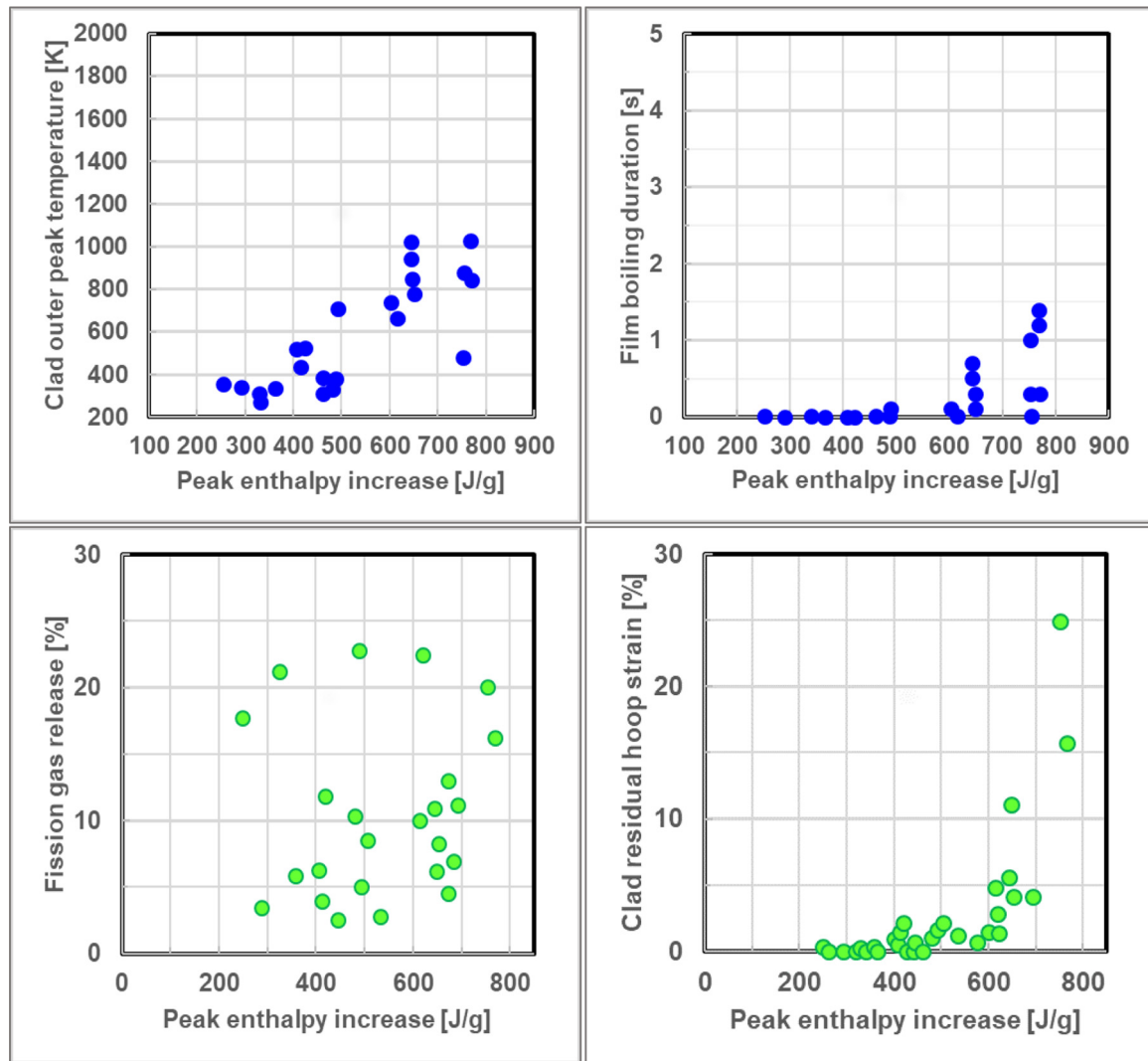


Fig. 1. NSRR RIA test data for PWR fuel rods. Evolution versus reevaluated peak enthalpy increase of clad outer surface peak temperature (top left), film boiling duration (top right), transient fission gas release (bottom left) and clad residual hoop strain (bottom right). Adapted from [7].

evolving contact conditions and axial pellet-clad locking when the gap is closed,

- the solving of the axial transient heat and mass balance equations for both sodium and water coolants. For NSRR stagnant water conditions under various pressure and temperature, the post-DNB stage is considered as will be detailed in Section 3.2,
- material laws describing the nonlinear mechanical behavior of irradiated claddings (Zircaloy-4, M5_{Framatome}³) based on the extensive PROMETRA program performed on irradiated cladding samples submitted to high strain rates and high temperatures [43–45]. Based on a unified viscoplastic formulation (only one type of inelastic deformation) with no stress threshold between the elastic and viscoplastic regimes, these constitutive laws (see formulation in [46] and [47] for Zircaloy-4 and M5_{Framatome} respectively) aim at describing the anisotropic mechanical behavior of the (up to about 6 annual cycles) irradiated cladding material within large temperature (up to 1100 °C for Zircaloy-4 and 820 °C for M5_{Framatome}) and strain rate ranges (up to 5 s⁻¹),
- the modeling of grain boundary cracking and of the associated FGR occurring in UO₂ and MOX fuels (helium release can also

be included in the simulation [48] since it can contribute to the increase of the fuel rod pressure). The onset and location of grain boundary cracking in the fuel pellet is given by the oxide fuel mechanical law that has been identified from compression tests performed on non-irradiated samples at high strain rates (up to 10⁻¹ /s) and temperatures (up to 1700 °C) [49,50]. The amount of fission gases that is released from the regions where grain boundaries are cracked depends mostly on the precise assessment of fission gas partitioning prior to the RIA test, whether in inter- or intragranular position, in bubbles or dissolved, for both restructured (HBS: High Burnup Structure) and non-restructured radial areas of the pellet. It is given by a dedicated model called CARACAS details of which can be found in [51]. An additional temperature criterion derived from annealing tests is considered with regard to FGR from the HBS [52],

- the modeling of gas flow induced by axial pressure gradients inside the rod free volumes (including pellet dishes/chamfers/cracks and pellet-cladding radial gap, with their respective temperature) as an alternative to the more classical model assuming instantaneous equilibrium of pressure inside these volumes. It is mainly based on the ideal gas law and Darcy's law for gas flow under pressure P gradients within the rod considered as a porous medium characterized by its

³ M5_{Framatome} is a trademark or registered trademark of Framatome or its affiliates, in the USA or other countries.

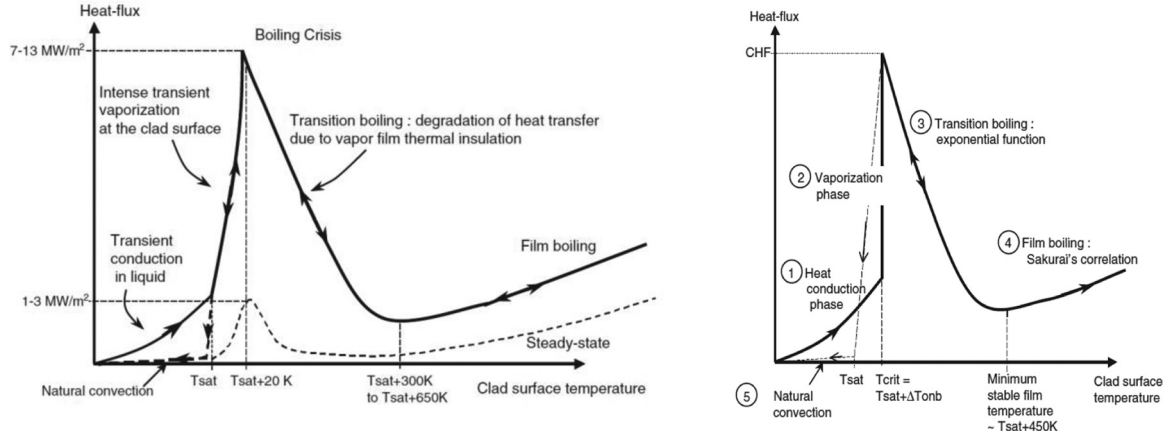


Fig. 2. Clad-to-water heat transfer (reproduced from [54]) Left: schematic view of the experimental boiling curve in the NSRR tests (compared to the one in steady state conditions). Right: heat transfer regimes considered in the model.

permeability K , following Eqn 1:

$$\frac{\partial}{\partial t} \left(\frac{\omega}{T} P \right) = \frac{d}{dz} \left(\frac{P}{T} \frac{K}{\mu} \frac{dP}{dz} \right) + R\dot{n} \quad (1)$$

where $\frac{\omega}{T}$ stands for the gas storage capacity taking account of the different rod free volumes with their respective temperature, R is the universal gas constant, μ the gas dynamic viscosity and $\dot{n}$ the amount of gas released. It allows modeling the kinetics of gas axial redistribution in the fuel rod during a RIA transient that could explain localized clad ballooning observed in some NSRR tests,

- the accounting for the thermal resistance of outer zirconia depending on its thickness through a modification of the clad-to-coolant transfer coefficient (the zirconia thickness is currently not explicitly meshed).

One important feature of ALCYONE is the absence of any user-dependent specific initialization of the variables prior to pulse-irradiation simulations since the latter start from the base irradiation conditions that the code itself computes.

Several criteria (on temperature, internal pressure, gas swelling strain and pellet-clad gap) are considered for the convergence of ALCYONE calculations. The time step can furthermore be automatically predicted and subdivided if need be during the calculation.

3.2. Boiling curves used in ALCYONE for water coolant and transient conditions

Solving the heat and mass balance equations requires the estimation of the heat flux between the clad outer surface and the water coolant. The clad-to-coolant heat flux is based on the boiling curves (*i.e.* heat flux versus clad temperature) proposed by Bessiron et al. [53,54] for PWR (15 MPa, 280 °C, 4 m/s) and NSRR (0.1 MPa, room temperature, stagnant liquid water) conditions. The influence of the high heating rates involved in RIA is taken in the boiling curves into account, which renders the heat flux – temperature relation different from the one considered in steady state calculations, as shown in Fig. 2-left. The transient boiling curve for the NSRR Room Temperature (RT) test conditions (0.1 MPa) is illustrated in Fig. 2-right. In ALCYONE, the heat flux derived from the different regimes is prescribed in the thermal calculation. An explicit time integration scheme is used with a strong constraint on the time step, in particular when the Critical Heat Flux (CHF) is reached. These developments have been tested successfully during the recent NEA RIA benchmark Phase II [55].

With regard to the simulation of NSRR RT tests, the correlations for stagnant liquid water conditions were derived by Bessiron from inverse analyses with the SCANAIR code [47,54,56] of NSRR tests performed on PWR pre-oxidized fresh fuel rods [59]. With T_{sat} denoting the saturation temperature, the transient boiling curve includes four different regimes (Fig. 2-right):

- heat conduction in the stagnant liquid water up to the critical temperature ($T_{sat} + 20$ K),
- vaporization of a 30 μm thick layer (water thickness threshold denoted w_{tt} in the following sections) of water at constant temperature ($T_{sat} + 20$ K). This regime was introduced to account for the impact of the energy deposition rate on the critical heat flux,
- transition and film boiling regimes are simulated with a heat flux that decreases exponentially with the clad temperature up to $T_{sat} + 450$ K and then asymptotically tends to the film boiling heat flux estimated by Sakurai for steady-state conditions Φ_{SSH}^* [11]; the application to fast transients led Bessiron to introduce a multiplying factor k_B to best fit the NSRR tests [54] so that the heat flux in the film boiling regime follows Eqn 2:

$$\Phi = (\Phi_{crit} - \Phi_{SSH}) e^{[-k(T_{clad} - T_{crit})^2]} + \Phi_{SSH} \quad (2)$$

where Φ_{crit} and T_{crit} denote the critical heat flux and temperature respectively, T_{clad} denotes the clad outer surface temperature, $k = 4.10^{-5} \text{ K}^{-2}$ and $\Phi_{SSH} = k_B \Phi_{SSH}^*$. In ALCYONE, $k_B = 5$,

- the rewetting phase is activated when the temperature of the minimum heat flux is reached ($T_{sat} + 450$ K). The heat flux is calculated according to the same three previous correlations. This approach differs from the one proposed by Bessiron et al. [54] but was found to have little impact on the results [55].

In case of ALCYONE pulse-irradiation simulations, an additional convergence criterion on the clad-to-coolant flux is considered.

4. ALCYONE simulations of selected NSRR tests

4.1. Assumptions and methodology

In the following simulations, 1D axisymmetric calculations are performed with ALCYONE. 30 axial segments are used for discretizing the full-length parent rod with a refinement at the location of the pulse-tested rodlet, discretized with axial slices of equal length (maximum 15 mm long). In the radial direction, the clad is discretized with 10 annular rings of equal thickness whereas the fuel

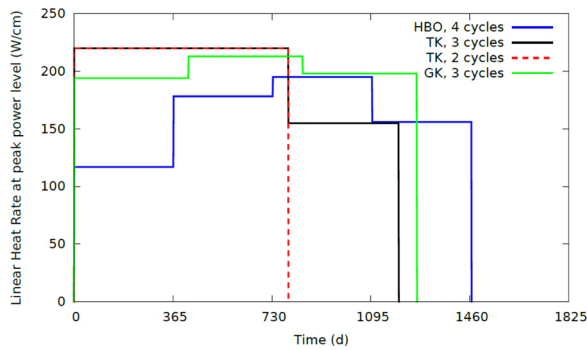


Fig. 3. Time evolution of linear heat rate at peak power level considered in ALCYONE calculations.

pellet mesh comprises 33 rings with the outer ones shorter to well capture the HBS formation.

To be performed, any ALCYONE calculation requires input data including parent fuel rod and rodlet manufacturing data as well as the parent rod base irradiation and the rodlet pulse irradiation power histories. With regard to the tests performed within JAEA NSRR programs, most of the data required for this study have been compiled from open-literature sources (see Appendix) and completed as follows when data were missing.

Concerning base irradiation of the parent rods:

- for test cases GK-1 and 2, simplified base irradiation histories have been built on the basis of average linear heat generation rate data and number of cycles;
- for test cases HBO and TK considered, simplified base irradiation Linear Heat Rate (LHR) histories have been built on the basis of data available for relevant (in terms of fuel and cladding materials and geometry) rods irradiated in Ohi-1 and Takahama-3 reactors (as were HBO and TK parent rods respectively) [57,58]. They aimed at leading to consistent average rodlet burnups and outer zirconia layer thickness prior to the pulse test considered. Thus, within each series, tests with similar burnup have been simulated with different base irradiation histories (depending on the elevation on the calculated fake parent rod) in case they had different outer zirconia thickness.

All the LHR histories considered during base irradiation are plotted in Fig. 3 (TK, 2 cycles, for cases TK-1 and 6; TK, 3 cycles for cases TK-4, 5, 8 and 9).

Concerning pulse irradiation conditions:

- power pulse histories were built from the VA-3 test whose detailed characteristics were available among the specifications of the 2013 phase I NEA benchmark [60]; the coupling factor was adjusted consistently with the peak fuel enthalpy increase achieved during the pulse test considered; the pulse FWHM was adjusted according to a time-scale expansion/contraction if needed (the Gaussian shape of the pulse is not changed);
- in case of lack of precise information about the axial power profile, it has been considered flat in ALCYONE simulations consistently with the LHR being almost uniform along the (short) pellet stack;
- for all test cases but GK-1 and 2, estimations of rodlet free volumes based on post-test measured RH-1 and RH-2 rodlet free volumes (4 cm³ and 1 cm³ respectively) [12] have been used in ALCYONE calculations. For test cases GK-1 and 2, the post-test measured free volume (3.9 cm³ and 3.6 cm³ respectively) [29] has been considered. No lower plenum has been taken into account. These choices have to be kept in mind since the free

volume is an essential parameter with respect to clad ballooning.

In a first set of ALCYONE calculations, pressure equilibrium in the rod free volumes is assumed, outer zirconia spallation is not considered. Complementary calculations accounting for zirconia spallation at starting time of the transient have been performed when it comes to comparing clad outer surface temperature results with measurements as will be detailed after.

4.2. Calculated average burnup, peak enthalpy increase and fission gas inventory

For the 11 tests selected, the ALCYONE calculated conditions characterizing both the end-of-base irradiation state (rodlet average burnup) and the transient pulse (enthalpy increase at peak power level, power evolution) are synthesized in Figs. 4 and 5, and compared to the values mentioned in the open literature. The wide ranges investigated for the burnup and for the mean enthalpy increase (between 250 and 750 J/g) as well as the small differences in pulse FWHM can easily be visualized.

ALCYONE calculated FGR during base irradiation and the intergranular fission gas inventory before pulse testing are listed in Table 2. Fission gas release that occurred during base irradiation is excluded from the calculated pre-test intergranular fission gas inventory, which is relatively constant (around 14–15%) as a consequence of the narrow burnup range of the fuel rodlets (38 GWd/tM – 51 GWd/tM). The base irradiation conditions lead to rather low calculated release fractions (less than 1%) globally consistent with the measured FGR during base irradiation (less than 0.5% for HBO and TK series). The pre-test intergranular fraction in the fuel is high, meaning that the potential for FGR during pulse test is important.

4.3. Calculated clad outer surface peak temperature and film boiling duration

Otherwise stated, the following figures present results relative to the slice located at the maximum linear heat rate.

The instrumentation device of the NSRR tests includes thermocouples spot-welded on the clad, placed at one or several axial locations. Consistently with the removal of the oxidation layer below spot-welded thermocouples [62], the calculated time evolution of the clad outer surface temperature is plotted in Fig. 6 assuming the spallation of the outer zirconia layer at starting time of the transient.

Fig. 7 shows the calculated evolution of both clad outer surface peak temperature (left-hand) and film boiling duration (right-hand) versus peak enthalpy increase that can be compared to the graphs of Fig. 1 (same x- and y-scales).

To better evidence the discrepancies, Fig. 8 compares ALCYONE calculated and measured values for both clad outer surface peak temperatures (left-hand) and film boiling durations (right-hand). At first sight, the calculated temperatures appear overestimated, especially when DNB occurs, but they are within the uncertainty of measurements (200 K – 300 K) due to some thermocouple fin effect [63]. Whether irradiation contributes to the overestimation is not obvious (Bessiron's boiling curve was established from tests performed on virgin samples [54]). The order of magnitude of the calculated clad outer surface peak temperatures is thus considered correct and the general trend of Fig. 1 recovered. ALCYONE calculations exhibit a steady increase of the clad outer surface peak temperature with peak enthalpy increase from a value of around 320 J/g. In Fig. 1, the temperature increase occurs a bit later, at an enthalpy increase of 400–450 J/g.

Similarly, ALCYONE calculations of the film boiling duration exhibit a steady increase with peak enthalpy increase starting from

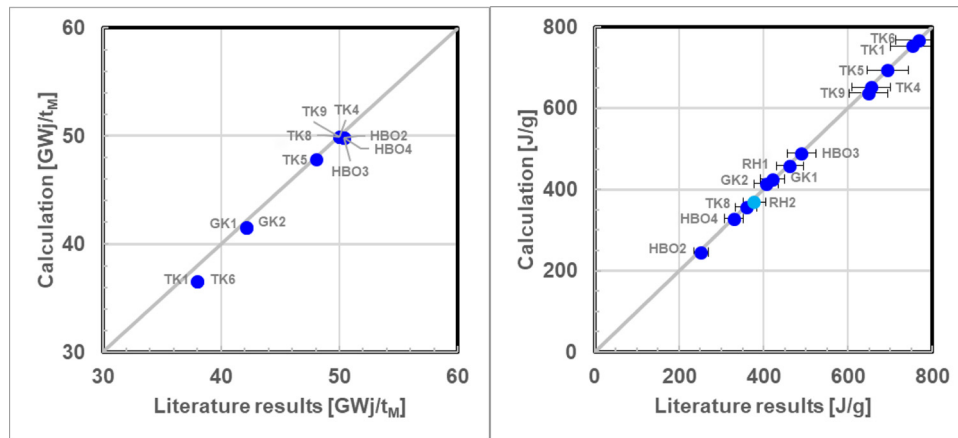


Fig. 4. ALCYONE calculated versus literature values of (left) burnup and (right) peak enthalpy increase.

Table 2

ALCYONE calculated burnup, FGR during base irradiation and pre-test rodlet intergranular fission gas inventory.

Test ID.	Burnup (GWd/tM)	FGR during base irradiation (%) ^a [measurements, see Appendix]	Pre-test intergranular fraction (%) ^b
GK-1	41.56	0.6 [0.76]	14.3
GK-2	41.56	0.6 [0.76]	14.3
HBO-2	51.11	0.9 [0.49]	14.5
HBO-3	50.82	0.9 [0.49]	14.2
HBO-4	50.35	0.9 [0.49]	14.1
TK-1	38.51	0.4 [< 0.4]	14.6
TK-4	50.82	0.9 [< 0.4]	15.3
TK-5	48.22	0.9 [< 0.4]	15.4
TK-6	38.77	0.5 [< 0.4]	15.6
TK-8	50.38	0.9 [< 0.4]	14.9
TK-9	50.39	0.9 [< 0.4]	14.6

^a of the total amount of gas created within the fuel pellets.

^b of the amount of gas retained within the fuel pellets after base irradiation.

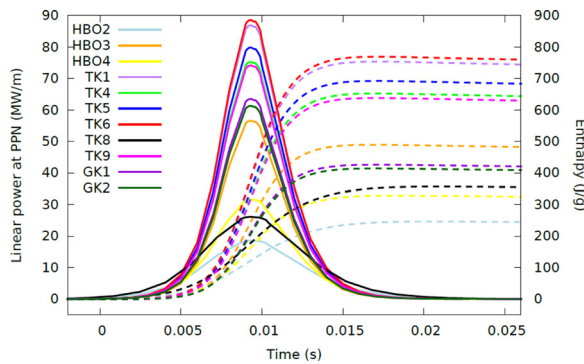


Fig. 5. Time evolution of linear heat rate (left axis, solid line) and enthalpy (right axis, dashed line) at peak power level (N.B. LHR curves are centered on the same time only for representation purpose).

around 320 J/g. Although overestimated, and whether irradiation contributes to the overestimation is here again not obvious, the order of magnitude of the calculated film boiling duration (0 to 2.5 s) given by the clad-to-coolant heat exchange model in ALCYONE is considered correct. Nevertheless, when compared with the general trend exhibited in Fig. 1, boiling occurrence seems overestimated which incidentally explains partly the overestimation of the clad outer surface peak temperature and questions the value of the critical heat flux considered.

Actually the clad outer surface peak temperature and film boiling duration have been determined from the models described in Section 3.2 that have a large part of empiricism and need to be updated, especially since the reevaluation of the fuel enthalpies (10 to 65%) that took place in 2014 [7,8]. From an experimental point

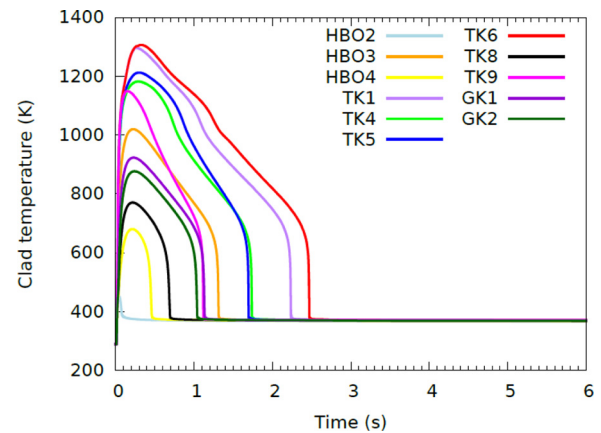


Fig. 6. ALCYONE calculated time evolution of the clad outer surface temperature assuming the outer zirconia is totally spalled at the starting time of the transient.

of view, Sugiyama reported for instance an increase by a factor of 4 of the film boiling duration induced by a variation of +50% in peak fuel enthalpy during tests performed on non-oxidized samples [59]. Given the current lack of experimental investigations on boiling heat transfer in RIA conditions [9], some readjustment of the heat transfer correlations sounds legitimate. It is the purpose of the next section.

4.4. Readjustment of the boiling curve parameters for NSRR tests

Sensitivity analyses have been performed on several parameters outlined in Section 3.2.

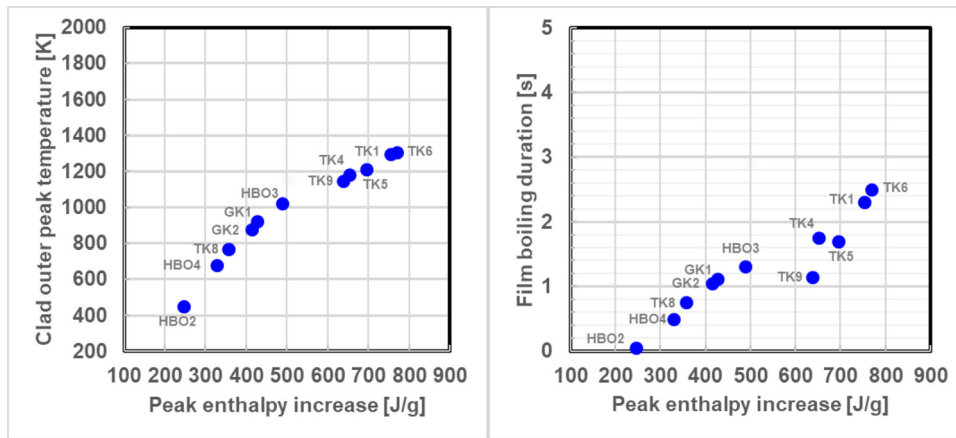


Fig. 7. ALCYONE calculated values of clad outer surface peak temperature (left) and film boiling duration (right) versus peak fuel enthalpy increase.

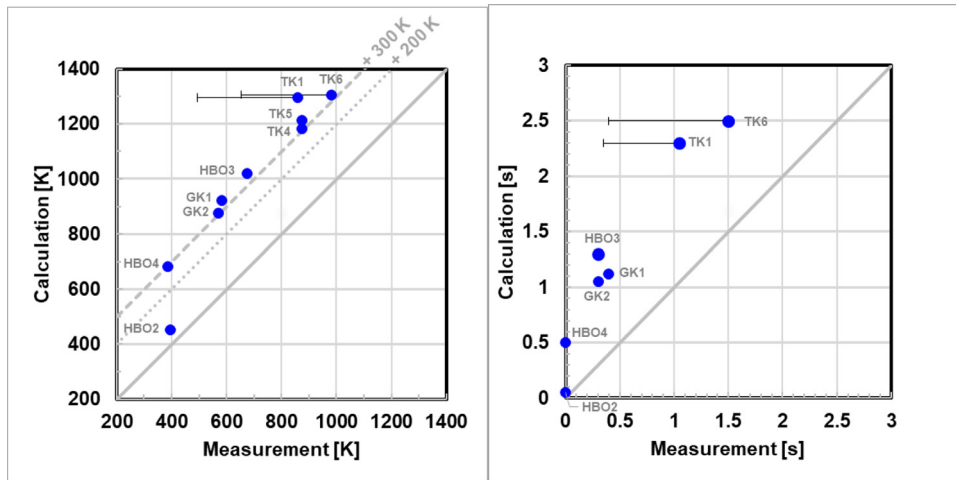


Fig. 8. ALCYONE calculated versus measured values of (left) clad outer surface peak temperature and (right) film boiling duration.

With regard to the correlations for NSRR RT tests, varying the water thickness that needs to be vaporized before the boiling crisis occurs denoted wtt had no obvious influence on the critical heat flux in the calculations, contrary to the heat flux at rewetting. The reference value of $30 \mu\text{m}$ has therefore been kept. The reference value of 4.10^{-5} K^{-2} for the decaying exponential parameter k has also been kept. Sensitivity analyses have shown that varying the parameter k_B around the reference value of 5 (discrete values 4 to 8 were tested) led in some case to a better prediction of film boiling durations and incidentally of peak clad temperatures. Given the known uncertainties on the clad temperature measurements (200 K – 300 K), parameter fitting on the film boiling duration has been favored. A value of 8 has finally been retained.

Fig. 9 shows ALCYONE calculated clad outer surface peak temperatures and film boiling durations versus measurements after readjustment of the boiling curves parameters. Compared to Fig. 8, it evidences the global improvement for all the tests considered. As can be seen, calculated and measured clad temperatures still differ by 100 K – 200 K depending on the test. The general trend is at an overestimation of the measured clad temperatures. This result is inherent to the measuring device which leads to underestimated clad temperatures by 200 K – 300 K, as discussed in reference [63]. Considering this systematic bias, the calculated temperatures can be considered satisfactory.

Fig.10 shows ALCYONE calculated clad outer surface peak temperatures and film boiling durations versus peak enthalpy increase after parameter fitting. The general trend when compared

to Fig. 7 is to lower temperatures and shorter film boiling durations. As the sensitivity analysis on parameter wtt did not show the expected influence on the value of critical heat flux, the over-estimation of film boiling initiation around 300 J/g has not been improved.

4.5. Calculated transient FGR

Now that the thermal heat transfer parameters have been optimized, the calculated FGR can be discussed. Transient fission gas release refers to the incremental fission gas release during the pulse irradiation, exclusive of fission gas release during base irradiation.

Fig.11 shows calculated transient fission gas release versus peak enthalpy increase. The graph can be compared to Fig.1 where all the measurements available from NSRR RT tests are plotted. Having the same axes and the same scales, the two figures can thus be compared. Fig.12 shows the comparison between ALCYONE calculation results and measurements.

As shown in Fig.1, the experimental dispersion is significant. Nevertheless, keeping apart the few test cases showing a FGR of $\sim 20\%$ independently of the enthalpy increase, the general trend is that FGR globally increases with enthalpy increase above 300 J/g. ALCYONE calculations exhibit a similar trend with a steady increase of FGR with enthalpy increase till stabilization around a level consistent with the pre-transient intergranular gas inventory (around 15%, see Table 2).

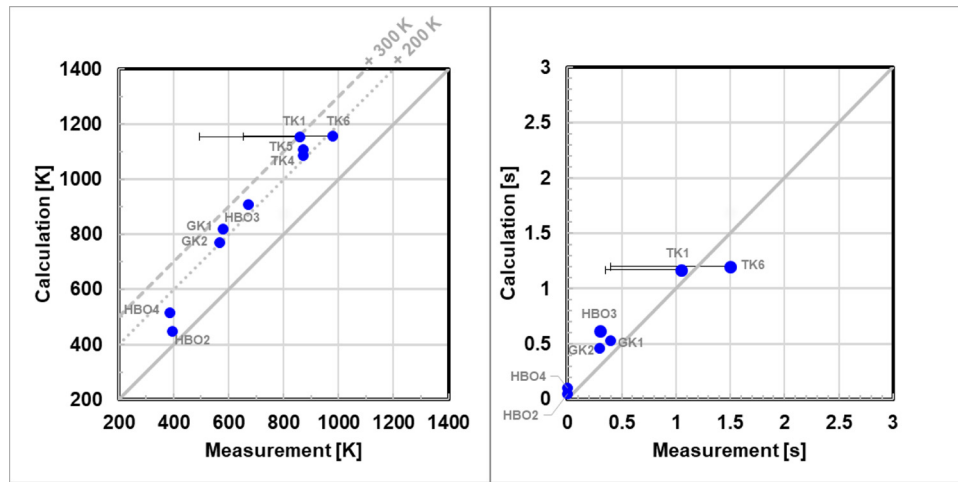


Fig. 9. ALCYONE calculated versus measured values of clad outer surface peak temperature (left) and film boiling duration (right) after parameter fitting on k_b .

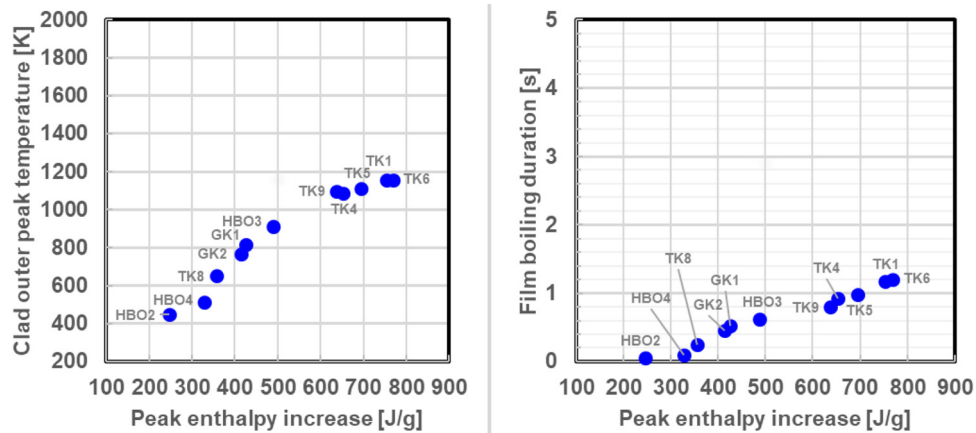


Fig. 10. ALCYONE calculated values of clad outer surface peak temperature (left) and film boiling duration (right) versus peak enthalpy increase after parameter fitting on k_b .

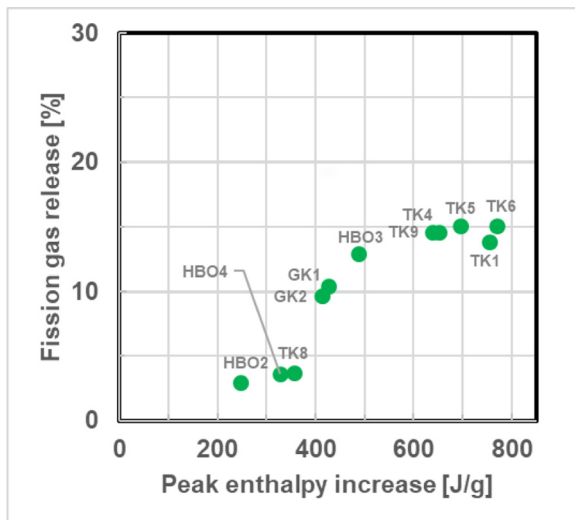


Fig. 11. ALCYONE calculated results of transient fission gas release versus peak fuel enthalpy increase.

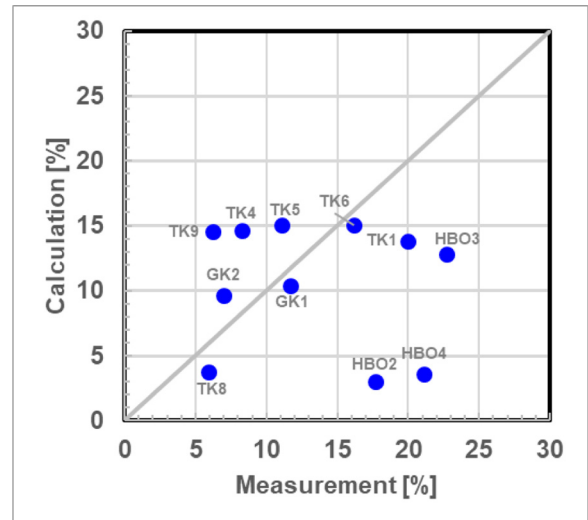


Fig. 12. ALCYONE calculated versus measured values of transient fission gas release.

Actually the pulse width is the main factor in the coupling used in ALCYONE between the fuel mechanical model and transient FGR. It determines the compressive stress level within the fuel pellet during the power increase part of the pulse which triggers grain

boundary cracking (and FGR) in ALCYONE. It seems thus not surprising for example that the less energetic TK-9 and HBO-3 tests and the more energetic TK-1 test have equivalent FGRs (the FWHM is identical). As pointed out in the CSNI report [9], the HBO tests (2 and 4) present rather unexpected high FGR (17 and 22%) con-

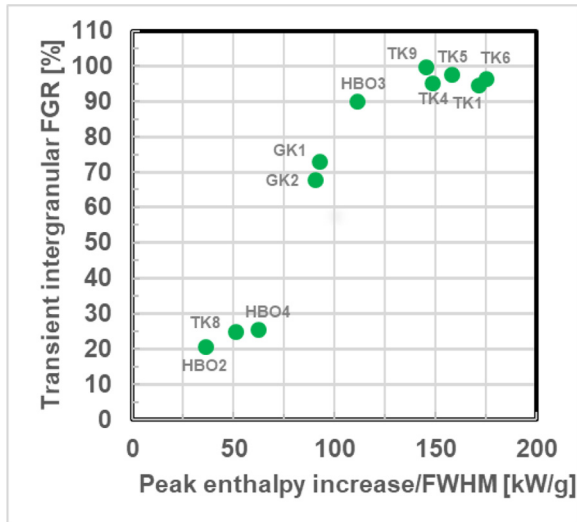


Fig. 13. ALCYONE calculated results of transient intergranular fission gas release versus ratio of peak fuel enthalpy increase to FWHM.

sidering their moderate fuel enthalpy increase during the tests and the somewhat larger FWHM (7 ms). In spite of an intergranular inventory prior to the test that is not negligible (18%), ALCYONE simulations give fairly small FGR for these two tests (less than 5%). This can be explained as follows.

Following recommendations made in the recently updated CSNI state-of-the-art report on fuel behavior under RIA conditions [9], Fig.13 shows the transient intergranular gas release (given by the ratio of the transient fission gas release to the pre-test intergranular fission gas fraction in the fuel, cf. Table 2) versus the ratio of peak fuel enthalpy increase to pulse width. This ratio was found to correlate better with FGR in the neural network study of NSRR tests by Koo et al. [41]. It is actually a measure of the pulse kinetics and hence of the thermomechanical stresses within the fuel pellets during the power increasing phase. An interesting trend in ALCYONE simulations can be observed, probably less scattered than the one obtained for transient FGR. Recalling that ALCYONE FGR is mostly driven by compressive stresses during the power increasing phase (and hence dependent on the pulse kinetics), this explains why the calculated FGR during the HBO tests are small. Obviously, another mechanism has to be taken into account during these tests if consistent FGRs are to be calculated. In the CSNI report [9], the high FGRs are related to the development of fine radial cracks at the HBO pellets periphery. This tensile induced grain boundary cracking is not considered in ALCYONE yet, nor is the coupling with the CARACAS model. The associated FGR is therefore not described. In theory, if intragranular gas migration occurs during the tests, as observed in the CABRI REP-Na campaign for the most energetic tests [61], the transient intergranular FGR can exceed 100% of the pre-test inventory (see figure 36 of [9]). It is not the case in the NSRR experiments since the FWHM is too small to allow sufficient time at high temperature for intragranular gas diffusion to the grain boundaries to proceed.

4.6. Calculated clad residual hoop strains during pulse tests

Fig.14 shows the evolution of the clad residual hoop strain versus peak enthalpy increase as calculated by ALCYONE. This graph can be compared to the measures of Fig.1 that include however more NSRR tests. As can be seen, the small database of 11 tests considered in this paper is sufficient to study the main trends observed in Fig.1. Fig.15 shows the comparison between ALCYONE calculation results and measurements of the clad residual hoop

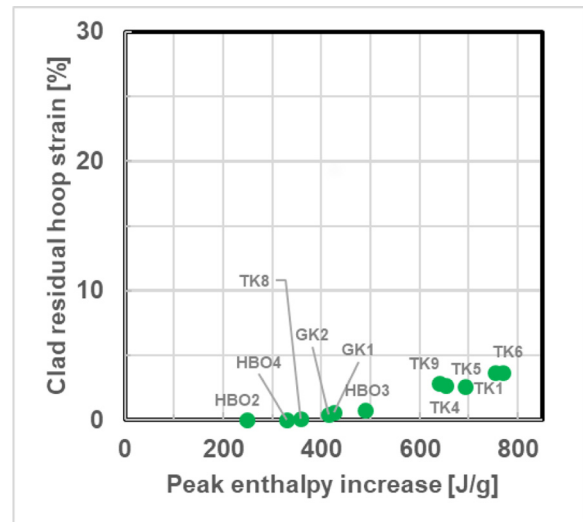


Fig. 14. ALCYONE calculated results of clad residual hoop strain versus peak fuel enthalpy increase.

strains generated during the NSRR pulse tests. In this figure, ALCYONE calculated values are compared to the measured axially mean clad residual hoop strains if available. If not available, the measured axially maximum value is considered and plotted in red.

The somewhat regular increase of residual hoop strain with peak enthalpy increase below ~ 600 J/g exhibited in Fig.1 is globally well reproduced by ALCYONE. Beyond 600 J/g, the impact of the high clad temperature on the material behavior (with a change of slope characteristic of clad ballooning) is not captured in ALCYONE calculations. Improvement is clearly needed on this point.

5. Investigation on the origin of clad ballooning during NSRR RIA tests

The NSRR tests that present significant ballooning are all from the TK-series. Simulations of these tests with ALCYONE showed that the FGR during the transient is correctly reproduced as well as the film boiling duration and the peak clad outer surface temperature. The underestimation of clad hoop strains during the transient cannot therefore be attributed to the modeling of thermal heat exchange and FGR in the code. Simulation trends during the TK-1 test are illustrated in Fig.16. They show that the FGR occurs quasi instantaneously during the power pulse (within 10 ms starting at around 0.242 s) and leads to a rise of the internal pressure till 0.43 MPa (y-axis scale in bar). At this time, the clad temperature is however not sufficient (less than 350 °C) to induce significant clad deformation. The pressure quickly stops increasing once clad deformation starts to increase. Obviously, the maximum pressure reached is not sufficient to induce more than 5% hoop strains during the transient.

It is with the abovementioned needed improvement for clad residual hoop strain simulation in mind that the model for delayed gas axial flow within the fuel rod has been implemented. A permeability of $10E-14$ m², consistent with the estimation of reference [64] based on gas flow rate during rod puncturing, has here been used.

Simulation trends during the TK-1 test are again illustrated in Fig.17 but this time considering that pressure equilibrium does not hold in the rod free volume. The lack of pressure equilibrium in the rod is illustrated in Fig.18. As shown in Fig.17, the kinetics of FGR is similar than in the previous simulation whereas the rod pressure evolution shows now a peak reaching 31 MPa. The residual clad hoop strain increases by a small amount (around 3.5%) but

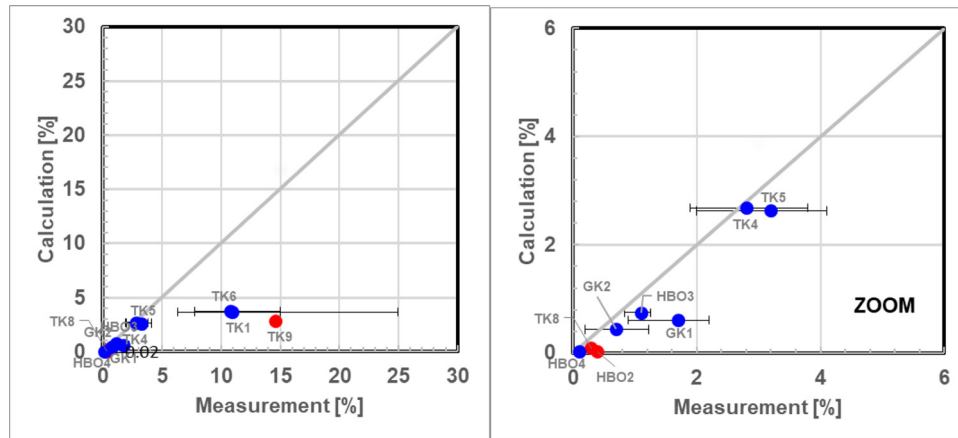


Fig. 15. ALCYONE calculated versus measured values of clad residual hoop strain.

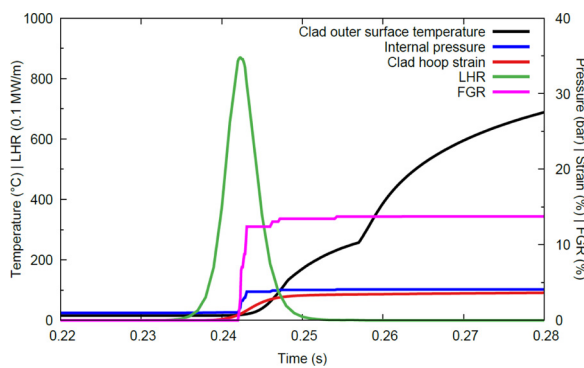


Fig. 16. Case TK-1. Pressure equilibrium reference calculation. Time evolution of various physical quantities of interest at peak power level (clad outer surface temperature; internal pressure; clad hoop strain; LHR; FGR).

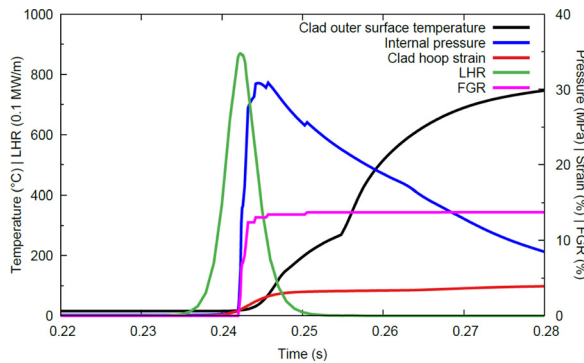


Fig. 17. Case TK-1. Calculation with delayed gas axial flow and no time-shift of FGR. Time evolution of various physical quantities of interest at peak power level (clad outer surface temperature; internal pressure; clad hoop strain; LHR; FGR).

remains far from the local 25% measured after the test. Additional simulations with a decreasing gas permeability in the fuel did not improve the results. The reason being that the clad temperature at the time of the overpressure is not sufficient to allow significant deformations to take place.

These simulations clearly highlight the need for a more suitable timing of fission gas release and rod internal pressure increase with respect to the clad temperature elevation. As the latter has already been optimized by a new parameter fitting of the boiling curve, some attempts have been made to artificially time-shift the fission gas release. This is illustrated in Fig. 19 on the TK-1 case showing the comparative time evolution of various physical quantities

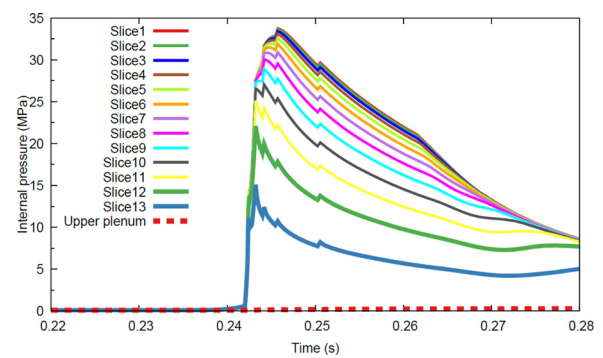


Fig. 18. Case TK-1. Calculation with delayed gas axial flow and no time-shift of FGR. Time evolution of internal pressure for each slice and the upper plenum of the rodlet.

ties at peak power level when assuming fission gas release occurs at pre-determined times after the start of the test, i.e., 0.3 s (subfigure a), 0.44 s (subfigure b) or 0.5 s (subfigure c). The corresponding time evolution of calculated clad hoop strains is plotted in the subfigure d.

Time-shifting the FGR leads to calculated clad outer surface temperature versus time curves more consistent with the measured ones [16]. In particular, they now exhibit a long plateau at high temperature during the film boiling phase (the same trend can be observed during the TK-6 test [54]). It also affects the film boiling duration, but marginally.

For each of the FGR times, a high peak of internal pressure is obtained, between 20 and 28 MPa (for an initial value of 0.1 MPa before pulse irradiation). This overpressure gives rise to a lasting reopening of the pellet-clad gap and to important clad hoop strains (up to 18%). Clad temperature has an impact on the internal pressure maximum level achieved in each case, as the ballooning tends to mitigate the over pressurization more rapidly when the clad temperature is higher.

This example shows that it is possible to achieve high levels of clad hoop strains, consistent with the TK-1 post-test measurements with residual strains of 25%, when considering a FGR time 0.44 s which actually corresponds to the occurrence of the peak clad outer surface temperature.

This result brings an alternative explanation to the one given in reference [16] where this ballooning was attributed to pronounced fission gas induced swelling of the fuel and pellet-clad gap was shown to remain closed. It is here considered that the obviously important level of pellet cracking with some pellet fragments stuck

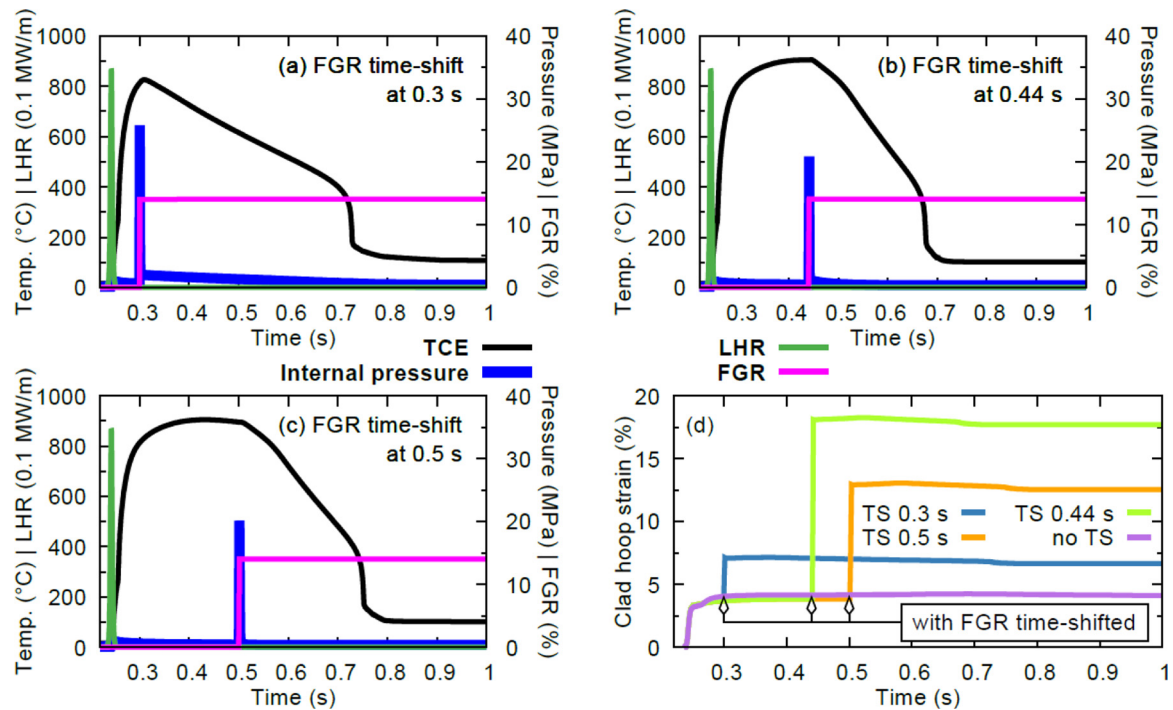


Fig. 19. Case TK-1. Calculation with delayed gas axial flow and time-shift of FGR. Illustration of the comparative time evolution of various physical quantities of interest at peak power level (clad outer surface temperature TCE, internal pressure, LHR and FGR on subfigures a to c; clad hoop strain on subfigure d).

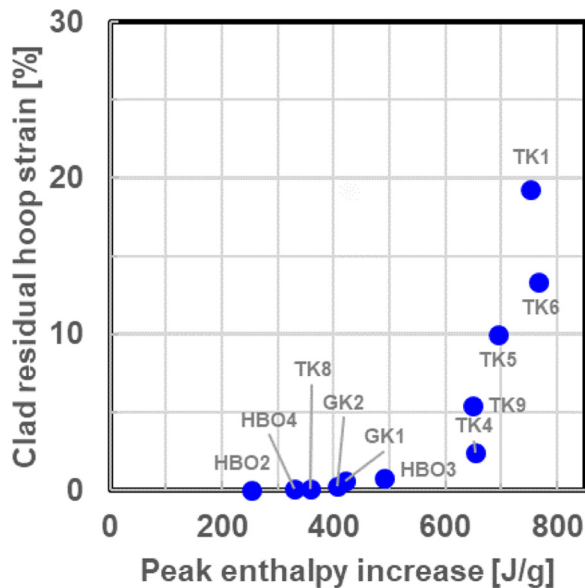


Fig. 20. ALCYONE calculated values of axially maximum clad residual hoop strain versus peak fuel enthalpy.

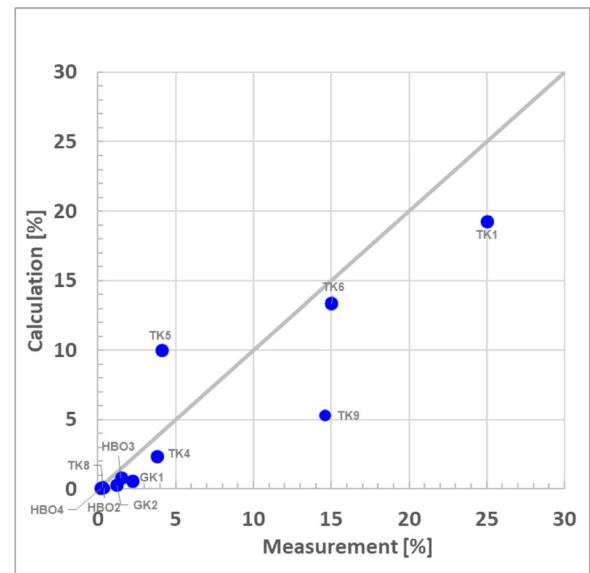


Fig. 21. ALCYONE calculated values versus measured values of axially maximum clad residual hoop strain.

to the clad inner surface might have led to the formation of some open gap space-shifted from the natural pellet-clad interface.

The same exercise as for TK-1, aiming at imposing fission gas release simultaneously with the peak clad outer surface temperature, has been repeated for each of the NSRR tests considered in this paper. Results in terms of calculated values of maximum residual hoop strain versus peak enthalpy increase are plotted in Fig. 20. The experimental change of slope around 600 J/g and the evolution at high enthalpy increase shown in Fig. 1 are well reproduced. Fig. 21 shows how much the comparison between ALCYONE calcu-

lated and measured values of maximum residual hoop strain has been improved with this new modeling.

From the curves of Fig. 19, one may note that the time-shifted fission gas release remains globally as instantaneous as the computed one in the reference simulation. Notwithstanding this rough approach, the whole set of results seems consistent and raises expectations that still improved calculated results could be achieved should progressive fission gas release be modeled and implemented in the ALCYONE code. This modeling should account for the gas propagation in open fuel voids (cracks, pores...) before they

could be released. The associated permeability of the fuel needs to be defined from appropriate experiments.

6. Conclusions and prospects

A selection of NSRR experiments were simulated with the ALCYONE fuel performance code to assess its capability in assessing clad ballooning occurring in fast transients performed in a water coolant.

This exercise showed that ALCYONE calculation results are in reasonable agreement with measurements performed during or after the NSRR tests. The rodlet thermal behavior and the transient FGR are satisfactorily captured despite some discrepancy for intermediate values of peak enthalpy increase and some overestimation of film boiling duration. The latter has been improved by a readjustment of the boiling curve parameters. As clad residual hoop strains were mostly underestimated especially at high enthalpy increase, a complementary work using the recently implemented gas flow model induced by axial pressure gradients has been proposed. It has shown that high levels of clad residual hoop strain could be achieved provided a proper timing of fission gas release, rod internal pressure increase and clad temperature elevation is used in the simulations. A progressive release of the fission gas from the fuel could therefore help modeling clad ballooning during the NSRR tests. Work aiming at modeling fission gas release kinetics in a more physical manner is underway including debates on what values of fuel permeability should be used in RIA conditions. In spite of these good results, clad hoop strain simulation might also be improved by taking the zirconium phase transformation and high temperature creep of Zircaloy-4 into account as they have been shown to be activated under simulated RIA thermo-mechanical conditions, cf. the range of temperatures of interest (840 °C – 1020 °C) investigated in the experimental and modeling work reported in [65,66] (even if the effects of very short time durations and irradiation-induced defects on resulting

clad strain are still pending questions). The modeling of these phenomena is currently not available in ALCYONE fuel performance code for RIA calculations. Nor is the modeling of the FGR associated with tensile induced (grain boundary) cracking, which may be based on recently published work [67,68]. For the work presented in this paper to be continued, key experimental and/or modeling research areas are thus at stake related to cladding deformation, fuel cracking, and how fuel cracking relates to fission gas release, the timing of which needs to be precisely assessed.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors are gratefully indebted to JAEA Udagawa-san for providing a list of data and references that helped completing the literature review.

The work described in the present paper has been performed in the continuity of many years of development of the ALCYONE fuel performance code at CEA with the constant financial and technical support of EDF and Framatome. The authors are thankful to the CEA colleagues for their fruitful contribution to the code development and the underlying physical modeling.

Appendix

Compiled details about PWR fuel specimens tested and tests performed in NSRR facility with stagnant water coolant [7,8,11–41].

M5_{Framatome} is a trademark or registered trademark of Framatome or its affiliates, in the USA or other countries.

	Test ID										
	GK-1	GK-2	HBO-2	HBO-3	HBO-4	TK-1	TK-4	TK-5	TK-6	TK-8	TK-9
Date of the test	03/12/1991	03/17/1992	03/25/1994	10/19/1994	01/24/1995	11/29/1996	02/04/1998	10/01/1998	10/07/1998	11/25/2000	12/25/2000
Power reactor for parent fuel rod irradiation	Genkai 1	Genkai 1	Ohi 1	Ohi 1	Ohi 1	Takahama 3	Takahama 3	Takahama 3	Takahama 3	Takahama 3	Takahama 3
Parent fuel rod manufacturer	Japan	Japan	Japan	Japan	Japan	Japan	Japan	Japan	Japan	Japan	Japan
Common parent rod with another test if any	n.m.	n.m.	MHI Ltd	MHI Ltd	MHI Ltd	MHI Ltd	MHI Ltd	MHI Ltd	MHI Ltd	NFI Ltd	NFI Ltd
Fuel rod design and type	–	–	–	–	–	TK-6	–	–	TK-1	–	–
	14 × 14	14 × 14	17 × 17	17 × 17	17 × 17	17 × 17	17 × 17	17 × 17	17 × 17	17 × 17	17 × 17
	PWR-UO ₂	PWR-UO ₂	PWR-UO ₂	PWR-UO ₂	PWR-UO ₂	PWR-UO ₂	PWR-UO ₂	PWR-UO ₂	PWR-UO ₂	PWR-UO ₂	PWR-UO ₂
Manufacturing data	Normal grain size	Normal grain size	Normal grain size	Normal grain size	Normal grain size	Normal grain size	Normal grain size	Normal grain size	Normal grain size	Normal grain size	Normal grain size
²³⁵ U initial enrichment (wt%)	3.4	3.4	3.2	3.2	3.2	4.1	4.1	4.1	4.1	4.1	4.1
Pellet outer diameter (mm)	9.29	9.29	8.19	8.19	8.19	8.19	8.19	8.19	8.19	8.19	8.19
Pellet height (mm)	15.2	15.2	13.5	13.5	13.5	10.0	10.0	10.0	10.0	10.0	10.0
Clad initial thickness (mm)	0.62	0.62	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57
Clad inner diameter ^a (mm)	9.48	9.48	8.36	8.36	8.36	8.36	8.36	8.36	8.36	8.36	8.36
Clad outer diameter ^a (mm)	10.72	10.72	9.5	9.5	9.5	9.5	9.5	9.5	9.5	9.5	9.5
Cladding alloy ^c	Zircaloy-4	Zircaloy-4	Zircaloy-4	Zircaloy-4	Zircaloy-4	Low-tin Zircaloy-4	Low-tin Zircaloy-4	Low-tin Zircaloy-4	Low-tin Zircaloy-4	Low-tin Zircaloy-4	Low-tin Zircaloy-4
Rodlet active fuel length (mm)	121	121	135	135	135	130	130	110	110	130–135	130–135
Sampling position from bottom (mm)	1618–1744	1435–1603									n.a.
[span from top]	[3/6]	[3/6]	[4/8]	[5/8]	[6/8]	[5]	[3]	[2]	[3]	[4]	
Rodlet fill gas	He	He	He	He	He	He	He	He	He	He	He
Rodlet fill gas pressure (MPa)	4.3	0.1	5.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	3
Rodlet free volume (cm ³)	3.9	3.6	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Base irradiation number of cycles	3	3	4	4	4	2	3	3	2	3	3
LHGR at last cycle (kW/m)	19.8	19.8	15.4	15.4	15.4	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
LHGR mean/max over base irradiation (kW/m)	20.2/n.m.	20.2/n.m.	15.5/19.5	15.5/19.5	15.5/19.5	19./20.5	19./20.5	19./20.5	19./20.5	19./20.5	19./20.5
Local burnup (GWd/tM)	42.1	42.1	50.4	50.4	50.4	38	50	48	38	50	50
Clad corrosion thickness (μm)	10	10	30 - 40	20 - 25	15 - 20	7	25	30	15	10	< 10
Clad hydrogen concentration ^d (wtppm)	n.a.	n.a.	152	148	89	50	125	163	100	n.a.	n.a.
FGR during base irradiation (%)	0.76	0.76	0.49	0.49	0.49	< 0.4	< 0.4	< 0.4	< 0.4	< 0.4	< 0.4
Capsule type	RT	RT	RT	RT	RT	RT	RT	RT	RT	RT	RT
Initial coolant temperature ^e (°C)	21	21	20	20	20	20	20	20	20	20	20
Coolant pressure (MPa)	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Pulse width (FWHM) (ms)	4.6	4.6	6.9	4.4	5.3	4.4	4.4	4.4	4.4	7.0	4.4
Energy deposit (J/g _{fuel})	506	490	213	397	280	674	523	544	669	351	527
Max. pellet stack axial elongation (mm)	1.28	0.72	0.6 ^p	n.a.	0.8	2.25	n.a.	n.a.	1.9	n.a.	n.a.
Max. cladding tube axial elongation (mm)	1.16	0.27	0.5	n.a.	0.6	2	n.a.	n.a.	1.6	n.a.	n.a.

(continued on next page)

(continued)

Peak fuel enthalpy ^b (J/g _{fuel})	421	407	252	490	330	753	654	694	768	359	649
Initial fuel enthalpy (at 20 °C) (J/g _{fuel})	0	0	0	0	0	0	0	0	0	0	0
Peak fuel enthalpy increase ^f (J/g _{fuel})	421	407	252	490	330	753	654	694	768	359	649
DNB detection	yes	yes	no	yes	no	yes	no	no	yes	no	no
Clad outer surface temperature ⁱ (K)											
max.	581	569	395	673	385	860	873	873	980	n.a.	n.a.
min.						493			653		
Film boiling duration (s)											
max.	0.4	0.3	–	0.3	–	1.05	–	–	1.5	–	–
min.				0.1		0.35			0.4		
Clad residual hoop strain ^g (%)											
max.	2.2	1.1	0.4	1.5	0.2	25	3.8	4.1	15.5	0.3	14.6
mean	1.7	0.7	n.a.	1.1	0.1	10.9	2.8	3.2	10.8		
min.	0.9	0.2	n.a.	0.5	0.	6.3	1.9	2.	7.8		
Transient FGR ^j (%)	11.7 ^h	7.0	17.7	22.7	21.1	20	8.3	11.1	16.2	5.9 ^h	6.2
Profilometry results	[29]	[29]		[27]	[27]	[16]	[16]	[16]	[16]		
Metallography results	[29]	[29]	[31]	[31,37,38]	[31]	[16]					
Temperature time evolution results	[29]	[29]	[38]	[54,27]		[16]			[54]		
Inner pressure time evolution results						[40]			[40]		[39]

n.m. = not measured or not mentioned; n.a. = not available. Normal grain size = around 10 µm.

LHGR = Linear Heat Generation Rate; DNB = Departure from Nucleate Boiling. RT stands for « Room-Temperature, Atmospheric-Pressure ».

^a nominal value.^b radial average.^c chemical composition – (standard) Zircaloy-4: Zr-1.5Sn-0.2Fe-0.1Cr by wt%; low-tin Zircaloy-4: Zr-1.3Sn-0.2Fe-0.1Cr by wt%.^d estimated relative uncertainty: less than 3% [8].^e estimated absolute uncertainty: less than 3 K [8].^f evaluation based on the total amount of fissile materials in a tested rod and computation by the neutron transport calculation model for the NSRR core. Estimated relative uncertainty: about 7% [6,8].^g azimuthally averaged.^h absolute uncertainty of 2%, accounting for variability of published data.ⁱ disregarding any smoothing of available curves; potential undermeasurement by 200–300 K [63].^j uncertainty multiplying factor of 1.6, accounting for variability of available data.^k highly fragmented end pellets if any may contribute a lot to measured transient FGR.

Clad outer surface peak temperature and film boiling uncertainty sources:

HBO-3 – Axial variability – Results for 2 TCs – DNB for 2 TCs – Film boiling only for TC located 40 mm below pellet stack center.**TK-1** (same burnup and enthalpy level as TK-6) – Axial variability – Results for 3 TCs – DNB for 3 TCs – Film boiling only for TCs located 42 mm (highest peak temperature) and 2 mm below pellet stack center.**TK-6** (same burnup and enthalpy level as TK-1) – Axial variability – Results for 3 TCs – Film boiling for 3 TCs – Highest peak temperature reached for TC located 42 mm above pellet stack center.

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